

Use-Case based exercises for programmers

1. Café Billing System

- **Scenario:** A café sells coffee at ₹120 per cup and tea at ₹80 per cup.
 - **Task:** Write a program that asks the customer how many coffees and teas they ordered, then prints the total bill.
 - **Hint:** Identify variables for quantities, unit prices, and total amount. Use arithmetic operators.
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2. Student Result Checker

- **Scenario:** A student's exam score is entered.
 - **Task:** Print "Pass" if marks ≥ 50 , otherwise "Fail".
 - **Hint:** Use relational operators and an if-else statement.
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3. Traffic Signal Decision

- **Scenario:** A traffic light shows either Red, Yellow, or Green.
 - **Task:** Based on the input, print "Stop", "Get Ready", or "Go".
 - **Hint:** Use conditional statements (if-else or switch).
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4. Discount Eligibility

- **Scenario:** A shop gives a 10% discount if the purchase amount is more than ₹1000.
 - **Task:** Ask the user for purchase amount and print final price after discount if eligible.
 - **Hint:** Use relational operators and arithmetic.
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5. Even Numbers Printer

- **Scenario:** A teacher wants to display all even numbers between 1 and 20.
 - **Task:** Write a program using a loop to print them.
 - **Hint:** Use for loop and modulus operator %.
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6. Multiplication Table Generator

- **Scenario:** A student wants to learn multiplication tables.
 - **Task:** Ask for a number and print its table up to 10.
 - **Hint:** Use for loop and multiplication.
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7. Voting Eligibility

- **Scenario:** Government portal checks if a person can vote.
 - **Task:** Input age, print "Eligible" if age ≥ 18 , else "Not Eligible".
 - **Hint:** Use conditional statements.
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8. Largest of Three Numbers

- **Scenario:** A math quiz asks the student to find the largest of three given numbers.
 - **Task:** Write a program to print the largest number.
 - **Hint:** Use nested if-else.
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9. Simple Login System

- **Scenario:** A system checks username and password.
 - **Task:** If both match predefined values, print "Login Successful", else "Access Denied".
 - **Hint:** Use logical operators (&&) and conditional statements.
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10. Sum of First N Natural Numbers

- **Scenario:** A student wants to calculate the sum of first N numbers.
 - **Task:** Input N and print the sum.
 - **Hint:** Use a loop and accumulator variable.
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11. Library Fine Calculator

- **Scenario:** A library charges ₹2 per day for late returns.
 - **Task:** Input the number of days late and calculate the fine. If fine > ₹50, print "Membership Suspended".
 - **Hint:** Variables → daysLate, fine. Use arithmetic + conditional.
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12. Temperature Converter

- **Scenario:** A weather app needs to convert Celsius to Fahrenheit.
 - **Task:** Input temperature in Celsius and print Fahrenheit. Formula: $F = (C * 9/5) + 32$.
 - **Hint:** Variables → celsius, fahrenheit. Use arithmetic.
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13. Bus Ticket Booking

- **Scenario:** A bus ticket costs ₹500. If passenger age < 12, they get 50% discount.
 - **Task:** Input age and print ticket price.
 - **Hint:** Variables → age, price. Use conditional.
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14. Bank Interest Calculator

- **Scenario:** A bank gives 5% annual interest.
 - **Task:** Input principal and years, calculate final amount. Formula: $\text{Amount} = \text{Principal} + (\text{Principal} * \text{Rate} * \text{Years}) / 100$.
 - **Hint:** Variables → principal, years, amount. Use arithmetic.
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15. Odd Number Printer

- **Scenario:** A math teacher wants all odd numbers between 1 and 30.
 - **Task:** Write a program using a loop to print them.
 - **Hint:** Use for loop and % operator.
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16. Grade Point Average (GPA) Calculator

- **Scenario:** A student has marks in 5 subjects.
 - **Task:** Input marks, calculate average, and print grade (A/B/C/F).
 - **Hint:** Variables → marks[], average. Use loop + conditional.
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17. Simple ATM Withdrawal

- **Scenario:** ATM allows withdrawal only if balance $\geq$ amount.
 - **Task:** Input balance and withdrawal amount. Print "Transaction Successful" or "Insufficient Balance".
 - **Hint:** Variables $\rightarrow$ balance, withdrawAmount. Use conditional.
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18. Electricity Bill Calculator

- **Scenario:** Electricity board charges ₹5 per unit. If units > 100 , extra surcharge of ₹200.
 - **Task:** Input units consumed and print bill.
 - **Hint:** Variables $\rightarrow$ units, bill. Use arithmetic + conditional.
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19. Palindrome Checker

- **Scenario:** A word is palindrome if it reads same forward and backward.
 - **Task:** Input a string and check if it's palindrome.
 - **Hint:** Variables $\rightarrow$ word, reversedWord. Use loop/string methods + conditional.
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20. Simple Quiz Game

- **Scenario:** A quiz asks 3 questions. Each correct answer gives 10 points.
 - **Task:** Input answers, calculate score, and print result.
 - **Hint:** Variables $\rightarrow$ score, answer. Use conditional + loop.
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21. Cinema Ticket Counter

- **Scenario:** A cinema charges ₹250 per ticket. If more than 5 tickets are booked, give 20% discount.
 - **Task:** Input number of tickets and print total cost.
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22. Water Bill Calculator

- **Scenario:** Water board charges ₹10 per unit. If usage > 500 units, add ₹1000 surcharge.
 - **Task:** Input units and print bill.
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23. BMI Calculator

- **Scenario:** Health app calculates $BMI = \text{weight} / (\text{height} * \text{height})$.
 - **Task:** Input weight (kg) and height (m), print BMI and category (Underweight/Normal/Overweight).
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24. Taxi Fare Estimator

- **Scenario:** Taxi charges ₹50 base fare + ₹20 per km.
 - **Task:** Input distance and print fare.
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25. Leap Year Checker

- **Scenario:** Calendar app checks if a year is leap year.
 - **Task:** Input year, print "Leap Year" or "Not Leap Year".
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26. Simple Calculator

- **Scenario:** User enters two numbers and an operator (+, -, *, /).
 - **Task:** Perform operation and print result.
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27. Hotel Room Booking

- **Scenario:** Hotel charges ₹2000 per night. If stay ≥ 7 nights, give 1 night free.
 - **Task:** Input nights and print total cost.
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28. Speed Limit Checker

- **Scenario:** Road speed limit is 60 km/h.
 - **Task:** Input speed, print "Over Speeding" if > 60 , else "Safe Driving".
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29. Currency Converter

- **Scenario:** Convert USD to INR (1 USD = ₹83).
 - **Task:** Input USD amount and print INR.
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30. Parking Lot Fee

- **Scenario:** Parking charges ₹30 per hour. If parked > 5 hours, flat ₹200.
 - **Task:** Input hours and print fee.
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31. Factorial Calculator

- **Scenario:** Math app calculates factorial of a number.
 - **Task:** Input N and print factorial using loop.
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32. Prime Number Checker

- **Scenario:** Teacher wants to check if a number is prime.
 - **Task:** Input number, print "Prime" or "Not Prime".
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33. Shopping Cart Total

- **Scenario:** A cart has 3 items with given prices.
 - **Task:** Input quantities, calculate total bill.
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34. Attendance Percentage

- **Scenario:** College requires 75% attendance.
 - **Task:** Input total classes and attended classes, print percentage and eligibility.
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35. Simple Interest Calculator

- **Scenario:** Formula: $SI = (P * R * T) / 100$.
 - **Task:** Input principal, rate, time, print interest.
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36. Reverse Number

- **Scenario:** A puzzle asks to reverse digits of a number.
 - **Task:** Input number, print reversed number.
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37. Armstrong Number Checker

- **Scenario:** A number is Armstrong if sum of cubes of digits = number.
 - **Task:** Input number, check Armstrong.
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38. Grade Distribution

- **Scenario:** School wants to classify marks: ≥ 90 A, ≥ 75 B, ≥ 50 C, else F.
 - **Task:** Input marks and print grade.
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39. Restaurant Bill Splitter

- **Scenario:** Friends share bill equally.
 - **Task:** Input total bill and number of friends, print share per person.
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40. Fibonacci Series Generator

- **Scenario:** Math app prints first N Fibonacci numbers.
 - **Task:** Input N and print series.
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41. Online Shopping Cart with Discount Codes

- **Scenario:** An e-commerce site allows customers to add 5 items to their cart. If the total exceeds ₹5000, apply a 15% discount. If a promo code "SAVE100" is entered, subtract ₹100 additionally.
 - **Task:** Input prices of 5 items and optional promo code, then print final bill.
 - **Concepts:** Arrays, loops, conditionals, string comparison.
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42. Student Marks Analyzer

- **Scenario:** A class has 10 students. Each student's marks are entered. The program should calculate:
 - Highest marks
 - Lowest marks
 - Average marks
 - **Task:** Input marks into an array and print results.
 - **Concepts:** Arrays, loops, min/max logic, arithmetic.
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43. Bank Transaction System

- **Scenario:** A bank account starts with ₹10,000. User can perform multiple transactions (deposit or withdraw). If withdrawal > balance, print "Insufficient Funds". After all transactions, print final balance.
 - **Task:** Input number of transactions, then process each one.
 - **Concepts:** Loops, conditionals, running totals.
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44. Word Frequency Counter

- **Scenario:** A text analyzer counts how many times each word appears in a sentence.

- **Task:** Input a sentence, split into words, and print frequency of each word.
 - **Concepts:** Strings, arrays, loops, conditionals.
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45. Simple Library Management

- **Scenario:** A library has 5 books stored in an array. User can:
 - Search for a book by title
 - Borrow a book (mark as unavailable)
 - Return a book (mark as available)
 - **Task:** Implement menu-driven program for these actions.
 - **Concepts:** Arrays, strings, loops, conditionals, menu system.
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46. Salary Slip Generator

- **Scenario:** A company pays a basic salary. If HRA = 20% of basic and DA = 10% of basic, calculate gross salary. If gross > ₹50,000, deduct 10% tax.
 - **Task:** Input basic salary, calculate gross and net salary.
 - **Concepts:** Arithmetic, conditionals, multiple variables.
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47. Number Guessing Game

- **Scenario:** System has a secret number (say 25). User keeps guessing until correct. Print "Too High" or "Too Low" until match.
 - **Task:** Implement guessing loop until correct number is entered.
 - **Concepts:** Loops, conditionals, user input validation.
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48. Simple Banking PIN Check

- **Scenario:** ATM allows 3 attempts to enter correct PIN. If correct, print "Access Granted", else "Card Blocked".
 - **Task:** Input PIN attempts, check validity.
 - **Concepts:** Loops, conditionals, counters.
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49. Restaurant Order System

- **Scenario:** A restaurant menu has 3 items:
 - 1 → Pizza ₹300
 - 2 → Burger ₹150
 - 3 → Juice ₹80
 - **Task:** User selects item number and quantity, program prints bill.
 - **Concepts:** Switch case, arithmetic, conditionals.
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50. Loan Eligibility Checker

- **Scenario:** Bank checks loan eligibility:
 - Age $\geq$ 21
 - Salary $\geq$ ₹25,000
 - No existing loan (boolean flag)
- **Task:** Input age, salary, loan status. Print “Eligible” or “Not Eligible”.
- **Concepts:** Relational + logical operators, conditionals.